

Sequences & Series

ANSWER KEY



Arithmetic sequences

- 1 For the arithmetic sequence with $a_1 = 6$ and common difference $d = 4$, find a_{12} .

$$a_{12} = a_1 + (12 - 1)d = 6 + 11(4) = 50.$$

- 2 For the arithmetic sequence with $a_1 = 20$ and $d = -3$, find a_{15} .

$$a_{15} = 20 + 14(-3) = 20 - 42 = -22.$$

- 3 A geometric sequence has $a_1 = 3$ and each term is 2 times the previous one. Find the smallest n for which $a_n = 384$.

$$a_n = 3 \cdot 2^{n-1}. \text{ Setting } 3 \cdot 2^{n-1} = 384 \text{ gives } 2^{n-1} = 128 = 2^7, \text{ so } n - 1 = 7 \text{ and } n = 8.$$

Arithmetic series (sum formula)

- 4 Find S_{12} for the arithmetic sequence with $a_1 = 6$, $d = 4$ (using $a_{12} = 50$ from above).

$$S_{12} = \frac{n(a_1 + a_n)}{2} = \frac{12(6 + 50)}{2} = 6(56) = 336.$$

- 5 Find S_8 for the arithmetic sequence with $a_1 = 2$, $d = 5$.

$$a_8 = 2 + 7(5) = 37. S_8 = \frac{8(2 + 37)}{2} = 4(39) = 156.$$

Geometric sequences

- 6 For the geometric sequence with $a_1 = 4$ and common ratio $r = 3$, find a_6 .

$$a_6 = a_1 r^5 = 4(3^5) = 4(243) = 972.$$

- 7 A geometric sequence begins 5, -10 , 20 , -40 , Find r and a_7 .

$$r = \frac{-10}{5} = -2. a_7 = 5(-2)^6 = 5(64) = 320.$$

Geometric series (finite)

- 8 Find S_6 for the geometric sequence with $a_1 = 4$, $r = 3$ (using $a_6 = 972$ from above).

$$S_6 = \frac{a_1(r^6 - 1)}{r - 1} = \frac{4(729 - 1)}{2} = \frac{4(728)}{2} = 1456.$$

- 9 Find S_5 for the geometric sequence $5, -10, 20, -40, \dots$

$$S_5 = \frac{5[(-2)^5 - 1]}{-2 - 1} = \frac{5(-32 - 1)}{-3} = \frac{5(-33)}{-3} = 55.$$

Infinite geometric series

- 10 Find the sum of the infinite geometric series $20 + \frac{20}{3} + \frac{20}{9} + \dots$.

$$a = 20, r = \frac{1}{3}, \text{ and } |r| < 1, \text{ so by Theorem 3.2 the sum is } \frac{20}{1 - \frac{1}{3}} = \frac{20}{\frac{2}{3}} = 30.$$

- 11 Find the sum of the infinite geometric series $9 - 3 + 1 - \frac{1}{3} + \dots$.

$$a = 9, r = -\frac{1}{3}, \text{ and } |r| < 1, \text{ so the sum is } \frac{9}{1 - (-\frac{1}{3})} = \frac{9}{\frac{4}{3}} = \frac{27}{4}.$$

- 12 Explain why the infinite geometric series with $a = 5, r = 2$ has no finite sum.

Theorem 3.2 requires $|r| < 1$ for the partial sums to converge. Here $|r| = 2 > 1$, so the terms grow without bound and the series diverges — it has no finite sum.

Multiple choice — easy

- 13 For the arithmetic sequence with $a_1 = -15$ and common difference $d = 8$, find a_{16} .

a. 113

b. 105 ✓

c. 97

d. 107

$$a_{16} = a_1 + (16 - 1)d = -15 + 120 = 105.$$

- 14 For the arithmetic sequence with $a_1 = 19$ and common difference $d = 6$, find a_{11} .

a. 73

b. 81

c. 79 ✓

d. 85

$$a_{11} = a_1 + (11 - 1)d = 19 + 60 = 79.$$

15 For the arithmetic sequence with $a_1 = 2$ and common difference $d = -4$, find a_{21} .

a. -74

b. -78 ✓

c. -76

d. -82

$$a_{21} = a_1 + (21 - 1)d = 2 - 80 = -78.$$

16 For the arithmetic sequence with $a_1 = -15$ and common difference $d = -5$, find a_{19} .

a. -110

b. -103

c. -100

d. -105 ✓

$$a_{19} = a_1 + (19 - 1)d = -15 - 90 = -105.$$

17 For the arithmetic sequence with $a_1 = 18$ and common difference $d = -3$, find a_{18} .

a. -36

b. -33 ✓

c. -31

d. -30

$$a_{18} = a_1 + (18 - 1)d = 18 - 51 = -33.$$

18 For the arithmetic sequence with $a_1 = 3$ and common difference $d = -5$, find a_{20} .

a. -97

b. -90

c. -92 ✓

d. -87

$$a_{20} = a_1 + (20 - 1)d = 3 - 95 = -92.$$

19 For the arithmetic sequence with $a_1 = 10$ and common difference $d = 7$, find a_{15} .

a. 108 ✓

b. 110

c. 101

d. 115

$$a_{15} = a_1 + (15 - 1)d = 10 + 98 = 108.$$

20 For the arithmetic sequence with $a_1 = -9$ and common difference $d = -1$, find a_{13} .

a. -19

b. -20

c. -21 ✓

d. -22

$$a_{13} = a_1 + (13 - 1)d = -9 - 12 = -21.$$

21 For the arithmetic sequence with $a_1 = 16$ and common difference $d = 8$, find a_{13} .

a. 114

b. 120

c. 104

d. 112 ✓

$$a_{13} = a_1 + (13 - 1)d = 16 + 96 = 112.$$

22 For the arithmetic sequence with $a_1 = 17$ and common difference $d = 5$, find a_{15} .

a. 82

b. 87 ✓

c. 92

d. 89

$$a_{15} = a_1 + (15 - 1)d = 17 + 70 = 87.$$

23 For the arithmetic sequence with $a_1 = -5$ and common difference $d = -8$, find a_{21} .

a. -163

b. -157

c. -165 ✓

d. -173

$$a_{21} = a_1 + (21 - 1)d = -5 - 160 = -165.$$

24 For the arithmetic sequence with $a_1 = -18$ and common difference $d = 6$, find a_{19} .

a. 92

b. 96

c. 90 ✓

d. 84

$$a_{19} = a_1 + (19 - 1)d = -18 + 108 = 90.$$

25 For the arithmetic sequence with $a_1 = 19$ and common difference $d = -6$, find a_{15} .

a. -71

b. -59

c. -63

d. -65 ✓

$$a_{15} = a_1 + (15 - 1)d = 19 - 84 = -65.$$

26 For the arithmetic sequence with $a_1 = 3$ and common difference $d = -4$, find a_{21} .

- a. -75
- b. -81
- c. -73
- d. -77 ✓

$$a_{21} = a_1 + (21 - 1)d = 3 - 80 = -77.$$

27 For the arithmetic sequence with $a_1 = -17$ and common difference $d = -4$, find a_{16} .

- a. -81
- b. -75
- c. -77 ✓
- d. -73

$$a_{16} = a_1 + (16 - 1)d = -17 - 60 = -77.$$

Multiple choice — medium

28 For the geometric sequence with $a_1 = 3$ and common ratio $r = -2$, find a_7 .

- a. 192 ✓
- b. -384
- c. 190
- d. 194

$$a_7 = a_1 r^6 = 3((-2)^6) = 3(64) = 192.$$

29 For the geometric sequence with $a_1 = 4$ and common ratio $r = -3$, find a_7 .

a. -8748

b. 2913

c. 2916 ✓

d. 2919

$$a_7 = a_1 r^6 = 4((-3)^6) = 4(729) = 2916.$$

30 For the geometric sequence with $a_1 = 4$ and common ratio $r = 3$, find a_7 .

a. 8748

b. 2913

c. 2916 ✓

d. 2919

$$a_7 = a_1 r^6 = 4((3)^6) = 4(729) = 2916.$$

31 For the geometric sequence with $a_1 = -1$ and common ratio $r = 2$, find a_6 .

a. -64

b. -32 ✓

c. -30

d. -34

$$a_6 = a_1 r^5 = -1((2)^5) = -1(32) = -32.$$

32 For the geometric sequence with $a_1 = -3$ and common ratio $r = 2$, find a_7 .

a. -192 ✓

b. -190

c. -194

d. -384

$$a_7 = a_1 r^6 = -3((2)^6) = -3(64) = -192.$$

33 For the geometric sequence with $a_1 = -1$ and common ratio $r = 2$, find a_4 .

a. -8 ✓

b. -10

c. -6

d. -16

$$a_4 = a_1 r^3 = -1((2)^3) = -1(8) = -8.$$

34 For the geometric sequence with $a_1 = 1$ and common ratio $r = -3$, find a_8 .

a. -2184

b. 6561

c. -2190

d. -2187 ✓

$$a_8 = a_1 r^7 = 1((-3)^7) = 1(-2187) = -2187.$$

35 For the geometric sequence with $a_1 = 1$ and common ratio $r = -2$, find a_6 .

a. -30

b. 64

c. -32 ✓

d. -34

$$a_6 = a_1 r^5 = 1((-2)^5) = 1(-32) = -32.$$

36 For the geometric sequence with $a_1 = 1$ and common ratio $r = -2$, find a_4 .

a. -8 ✓

b. -10

c. 16

d. -6

$$a_4 = a_1 r^3 = 1((-2)^3) = 1(-8) = -8.$$

37 For the geometric sequence with $a_1 = -3$ and common ratio $r = -3$, find a_6 .

a. 726

b. 729 ✓

c. 732

d. -2187

$$a_6 = a_1 r^5 = -3((-3)^5) = -3(-243) = 729.$$

38 For the geometric sequence with $a_1 = -3$ and common ratio $r = 2$, find a_8 .

a. -386

b. -384 ✓

c. -382

d. -768

$$a_8 = a_1 r^7 = -3((2)^7) = -3(128) = -384.$$

39 For the geometric sequence with $a_1 = 1$ and common ratio $r = 3$, find a_6 .

a. 240

b. 246

c. 243 ✓

d. 729

$$a_6 = a_1 r^5 = 1((3)^5) = 1(243) = 243.$$

40 For the geometric sequence with $a_1 = 2$ and common ratio $r = 3$, find a_6 .

a. 483

b. 1458

c. 486 ✓

d. 489

$$a_6 = a_1 r^5 = 2((3)^5) = 2(243) = 486.$$

41 For the geometric sequence with $a_1 = 3$ and common ratio $r = -3$, find a_6 .

a. -726

b. -732

c. -729 ✓

d. 2187

$$a_6 = a_1 r^5 = 3((-3)^5) = 3(-243) = -729.$$

42 For the geometric sequence with $a_1 = 4$ and common ratio $r = -2$, find a_5 .

a. -128

b. 64 ✓

c. 62

d. 66

$$a_5 = a_1 r^4 = 4((-2)^4) = 4(16) = 64.$$

Multiple choice — hard

43 Find S_{13} for the arithmetic sequence with $a_1 = -8$, $d = 1$.

a. -27

b. -26 ✓

c. -13

d. -4

$$a_{13} = a_1 + (13 - 1)d = -8 + 12 = 4. S_{13} = \frac{n(a_1 + a_n)}{2} = \frac{13(-8 + 4)}{2} = \frac{13(-4)}{2} = -26.$$

44 Find S_{13} for the arithmetic sequence with $a_1 = 10$, $d = 8$.

a. 116

b. 754 ✓

c. 746

d. 767

$$a_{13} = a_1 + (13 - 1)d = 10 + 96 = 106. S_{13} = \frac{n(a_1 + a_n)}{2} = \frac{13(10 + 106)}{2} = \frac{13(116)}{2} = 754.$$

45 Find S_{12} for the arithmetic sequence with $a_1 = 2$, $d = 9$.

a. 609

b. 630

c. 618 ✓

d. 103

$$a_{12} = a_1 + (12 - 1)d = 2 + 99 = 101. S_{12} = \frac{n(a_1 + a_n)}{2} = \frac{12(2 + 101)}{2} = \frac{12(103)}{2} = 618.$$

46 Find S_{12} for the arithmetic sequence with $a_1 = -1$, $d = 4$.

a. 42

b. 252 ✓

c. 248

d. 264

$$a_{12} = a_1 + (12 - 1)d = -1 + 44 = 43. S_{12} = \frac{n(a_1 + a_n)}{2} = \frac{12(-1 + 43)}{2} = \frac{12(42)}{2} = 252.$$

47 Find S_{12} for the arithmetic sequence with $a_1 = 7$, $d = -9$.

a. -510 ✓

b. -85

c. -501

d. -498

$$a_{12} = a_1 + (12 - 1)d = 7 - 99 = -92. S_{12} = \frac{n(a_1 + a_n)}{2} = \frac{12(7 - 92)}{2} = \frac{12(-85)}{2} = -510.$$

48 Find S_4 for the geometric sequence with $a_1 = 4$, $r = 3$.

a. 163

b. 157

c. 108

d. 160 ✓

$$S_4 = \frac{a_1(r^4 - 1)}{r - 1} = \frac{4(81 - 1)}{3 - 1} = \frac{320}{2} = 160.$$

49 Find S_4 for the geometric sequence with $a_1 = 2$, $r = -2$.

a. -10 ✓

b. -16

c. -8

d. -12

$$S_4 = \frac{a_1(r^4 - 1)}{r - 1} = \frac{2(16 - 1)}{-2 - 1} = \frac{30}{-3} = -10.$$

50 Find S_4 for the geometric sequence with $a_1 = 4$, $r = -3$.

a. -77

b. -108

c. -80 ✓

d. -83

$$S_4 = \frac{a_1(r^4 - 1)}{r - 1} = \frac{4(81 - 1)}{-3 - 1} = \frac{320}{-4} = -80.$$

51 Find S_6 for the geometric sequence with $a_1 = 6$, $r = -2$.

a. -126 ✓

b. -192

c. -124

d. -128

$$S_6 = \frac{a_1(r^6 - 1)}{r - 1} = \frac{6(64 - 1)}{-2 - 1} = \frac{378}{-3} = -126.$$

52 Find S_6 for the geometric sequence with $a_1 = -3$, $r = 3$.

a. -1095

b. -1092 ✓

c. -1089

d. -729

$$S_6 = \frac{a_1(r^6 - 1)}{r - 1} = \frac{-3(729 - 1)}{3 - 1} = \frac{-2184}{2} = -1092.$$

53 Find the sum of the infinite geometric series with $a_1 = 13$ and $r = \frac{1}{2}$.

a. 26 ✓

b. $\frac{26}{3}$

c. $\frac{13}{2}$

d. 27

$$\text{Since } |r| < 1: S = \frac{a_1}{1 - r} = \frac{13}{1/2} = 26.$$

54 Find the sum of the infinite geometric series with $a_1 = 7$ and $r = \frac{2}{3}$.

a. 21 ✓

b. $\frac{7}{3}$

c. $\frac{21}{5}$

d. $\frac{21}{2}$

$$\text{Since } |r| < 1: S = \frac{a_1}{1-r} = \frac{7}{1/3} = 21.$$

55 Find the sum of the infinite geometric series with $a_1 = 4$ and $r = -\frac{1}{2}$.

a. 8

b. $\frac{8}{3}$ ✓

c. 6

d. -8

$$\text{Since } |r| < 1: S = \frac{a_1}{1-r} = \frac{4}{3/2} = \frac{8}{3}.$$

56 Find the sum of the infinite geometric series with $a_1 = 9$ and $r = \frac{2}{3}$.

a. 27 ✓

b. 3

c. $\frac{27}{5}$

d. $\frac{27}{2}$

$$\text{Since } |r| < 1: S = \frac{a_1}{1-r} = \frac{9}{1/3} = 27.$$

57 Find the sum of the infinite geometric series with $a_1 = 18$ and $r = \frac{1}{5}$.

a. 90

b. $\frac{45}{2}$ ✓

c. 15

d. $\frac{72}{5}$

$$\text{Since } |r| < 1: S = \frac{a_1}{1-r} = \frac{18}{4/5} = \frac{45}{2}.$$